Analisi Diffusione COVID-19

Il committente richiede di avere un report su casi e vaccinazioni in diverse aree del mondo; a tal fine, richiede di utilizzare il dataset, curato da Our World in Data, all'indirizzo https://github.com/owid/covid-19-data/tree/master/public/data alla voce "Download our complete COVID-19 dataset" scaricare il dataset nel formato che si preferisce.

1. Si richiede di verificare le dimensioni del dataset e i relativi metadati

In [1]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

In [71]:
```
# Importo le librerie
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [72]:
```
# Importo il dataset e lo definisco come DataFrame
df_covid = pd.read_csv("/content/drive/MyDrive/EPICODE/M4_PYTHON/Prova_FIn

# Avvio l'EDA
# Stampo le prime e le ultime 5 righe
df_covid
```

Out[72]:

	iso_code	continent	location	date	total_cases	new_cases	new_cases_
0	AFG	Asia	Afghanistan	2020-01-05	0.0	0.0	
1	AFG	Asia	Afghanistan	2020-01-06	0.0	0.0	
2	AFG	Asia	Afghanistan	2020-01-07	0.0	0.0	
3	AFG	Asia	Afghanistan	2020-01-08	0.0	0.0	
4	AFG	Asia	Afghanistan	2020-01-09	0.0	0.0	
...	...	...	...	...	...	...	
429430	ZWE	Africa	Zimbabwe	2024-07-31	266386.0	0.0	
429431	ZWE	Africa	Zimbabwe	2024-08-01	266386.0	0.0	

429432	ZWE	Africa	Zimbabwe	2024-08-02	266386.0	0.0
429433	ZWE	Africa	Zimbabwe	2024-08-03	266386.0	0.0
429434	ZWE	Africa	Zimbabwe	2024-08-04	266386.0	0.0

429435 rows × 67 columns

In [73]:

```
#EDA - Verifico le dimensioni e i metadati
df_covid.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 429435 entries, 0 to 429434
Data columns (total 67 columns):
 #   Column                                   Non-Null Count   Dtype  
---  ------                                   --------------   -----  
 0   iso_code                                 429435 non-null  object 
 1   continent                                402910 non-null  object 
 2   location                                 429435 non-null  object 
 3   date                                     429435 non-null  object 
 4   total_cases                              411804 non-null  float64
 5   new_cases                                410159 non-null  float64
 6   new_cases_smoothed                       408929 non-null  float64
 7   total_deaths                             411804 non-null  float64
 8   new_deaths                               410608 non-null  float64
 9   new_deaths_smoothed                      409378 non-null  float64
 10  total_cases_per_million                  411804 non-null  float64
 11  new_cases_per_million                    410159 non-null  float64
 12  new_cases_smoothed_per_million           408929 non-null  float64
 13  total_deaths_per_million                 411804 non-null  float64
 14  new_deaths_per_million                   410608 non-null  float64
 15  new_deaths_smoothed_per_million          409378 non-null  float64
 16  reproduction_rate                        184817 non-null  float64
 17  icu_patients                             39116 non-null   float64
 18  icu_patients_per_million                 39116 non-null   float64
 19  hosp_patients                            40656 non-null   float64
 20  hosp_patients_per_million                40656 non-null   float64
 21  weekly_icu_admissions                    10993 non-null   float64
 22  weekly_icu_admissions_per_million        10993 non-null   float64
 23  weekly_hosp_admissions                   24497 non-null   float64
 24  weekly_hosp_admissions_per_million       24497 non-null   float64
 25  total_tests                              79387 non-null   float64
 26  new_tests                                75403 non-null   float64
 27  total_tests_per_thousand                 79387 non-null   float64
 28  new_tests_per_thousand                   75403 non-null   float64
 29  new_tests_smoothed                       103965 non-null  float64
 30  new_tests_smoothed_per_thousand          103965 non-null  float64
 31  positive_rate                            95927 non-null   float64
 32  tests_per_case                           94348 non-null   float64
 33  tests_units                              106788 non-null  object 
 34  total_vaccinations                       85417 non-null   float64
 35  people_vaccinated                        81132 non-null   float64
 36  people_fully_vaccinated                  78061 non-null   float64
 37  total_boosters                           53600 non-null   float64
 38  new_vaccinations                         70971 non-null   float64
 39  new_vaccinations_smoothed                195029 non-null  float64
 40  total_vaccinations_per_hundred           85417 non-null   float64
 41  people_vaccinated_per_hundred            81132 non-null   float64
 42  people_fully_vaccinated_per_hundred      78061 non-null   float64
```

```
 43  total_boosters_per_hundred                 53600 non-null   float64
 44  new_vaccinations_smoothed_per_million      195029 non-null  float64
 45  new_people_vaccinated_smoothed             192177 non-null  float64
 46  new_people_vaccinated_smoothed_per_hundred 192177 non-null  float64
 47  stringency_index                           196190 non-null  float64
 48  population_density                         360492 non-null  float64
 49  median_age                                 334663 non-null  float64
 50  aged_65_older                              323270 non-null  float64
 51  aged_70_older                              331315 non-null  float64
 52  gdp_per_capita                             328292 non-null  float64
 53  extreme_poverty                            211996 non-null  float64
 54  cardiovasc_death_rate                      328865 non-null  float64
 55  diabetes_prevalence                        345911 non-null  float64
 56  female_smokers                             247165 non-null  float64
 57  male_smokers                               243817 non-null  float64
 58  handwashing_facilities                     161741 non-null  float64
 59  hospital_beds_per_thousand                 290689 non-null  float64
 60  life_expectancy                            390299 non-null  float64
 61  human_development_index                    319127 non-null  float64
 62  population                                 429435 non-null  int64
 63  excess_mortality_cumulative_absolute       13411 non-null   float64
 64  excess_mortality_cumulative                13411 non-null   float64
 65  excess_mortality                           13411 non-null   float64
 66  excess_mortality_cumulative_per_million    13411 non-null   float64
dtypes: float64(61), int64(1), object(5)
memory usage: 219.5+ MB
```

Il dataset contiene 429435 righe e 67 colonne

In [74]:

```python
#EDA - Converto in datetime la colonna date
df_covid['date'] = pd.to_datetime(df_covid['date'])
display(df_covid['date'])
print("\n")
df_covid.info() #verifico l'effettiva trasformazione in datetime
```

	date
0	2020-01-05
1	2020-01-06
2	2020-01-07
3	2020-01-08
4	2020-01-09
...	...
429430	2024-07-31
429431	2024-08-01
429432	2024-08-02
429433	2024-08-03
429434	2024-08-04

429435 rows × 1 columns

dtype: datetime64[ns]

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 429435 entries, 0 to 429434
Data columns (total 67 columns):
 #   Column                                      Non-Null Count   Dtype
---  ------                                      --------------   -----
 0   iso_code                                    429435 non-null  object
 1   continent                                   402910 non-null  object
 2   location                                    429435 non-null  object
 3   date                                        429435 non-null  datetime64
[ns]
 4   total_cases                                 411804 non-null  float64
 5   new_cases                                   410159 non-null  float64
 6   new_cases_smoothed                          408929 non-null  float64
 7   total_deaths                                411804 non-null  float64
 8   new_deaths                                  410608 non-null  float64
 9   new_deaths_smoothed                         409378 non-null  float64
 10  total_cases_per_million                     411804 non-null  float64
 11  new_cases_per_million                       410159 non-null  float64
 12  new_cases_smoothed_per_million              408929 non-null  float64
 13  total_deaths_per_million                    411804 non-null  float64
 14  new_deaths_per_million                      410608 non-null  float64
 15  new_deaths_smoothed_per_million             409378 non-null  float64
 16  reproduction_rate                           184817 non-null  float64
 17  icu_patients                                39116 non-null   float64
 18  icu_patients_per_million                    39116 non-null   float64
 19  hosp_patients                               40656 non-null   float64
 20  hosp_patients_per_million                   40656 non-null   float64
 21  weekly_icu_admissions                       10993 non-null   float64
 22  weekly_icu_admissions_per_million           10993 non-null   float64
 23  weekly_hosp_admissions                      24497 non-null   float64
 24  weekly_hosp_admissions_per_million          24497 non-null   float64
 25  total_tests                                 79387 non-null   float64
 26  new_tests                                   75403 non-null   float64
 27  total_tests_per_thousand                    79387 non-null   float64
 28  new_tests_per_thousand                      75403 non-null   float64
 29  new_tests_smoothed                          103965 non-null  float64
 30  new_tests_smoothed_per_thousand             103965 non-null  float64
 31  positive_rate                               95927 non-null   float64
 32  tests_per_case                              94348 non-null   float64
 33  tests_units                                 106788 non-null  object
 34  total_vaccinations                          85417 non-null   float64
 35  people_vaccinated                           81132 non-null   float64
 36  people_fully_vaccinated                     78061 non-null   float64
 37  total_boosters                              53600 non-null   float64
 38  new_vaccinations                            70971 non-null   float64
 39  new_vaccinations_smoothed                   195029 non-null  float64
 40  total_vaccinations_per_hundred              85417 non-null   float64
 41  people_vaccinated_per_hundred               81132 non-null   float64
 42  people_fully_vaccinated_per_hundred         78061 non-null   float64
 43  total_boosters_per_hundred                  53600 non-null   float64
 44  new_vaccinations_smoothed_per_million       195029 non-null  float64
 45  new_people_vaccinated_smoothed              192177 non-null  float64
 46  new_people_vaccinated_smoothed_per_hundred  192177 non-null  float64
 47  stringency_index                            196190 non-null  float64
 48  population_density                          360492 non-null  float64
 49  median_age                                  334663 non-null  float64
 50  aged_65_older                               323270 non-null  float64
 51  aged_70_older                               331315 non-null  float64
 52  gdp_per_capita                              328292 non-null  float64
 53  extreme_poverty                             211996 non-null  float64
 54  cardiovasc_death_rate                       328865 non-null  float64
 55  diabetes_prevalence                         345911 non-null  float64
 56  female_smokers                              247165 non-null  float64
 57  male_smokers                                243817 non-null  float64
```

```
 57  male_smokers                             245817 non-null  float64
 58  handwashing_facilities                   161741 non-null  float64
 59  hospital_beds_per_thousand               290689 non-null  float64
 60  life_expectancy                          390299 non-null  float64
 61  human_development_index                  319127 non-null  float64
 62  population                               429435 non-null  int64
 63  excess_mortality_cumulative_absolute     13411 non-null   float64
 64  excess_mortality_cumulative              13411 non-null   float64
 65  excess_mortality                         13411 non-null   float64
 66  excess_mortality_cumulative_per_million  13411 non-null   float64
dtypes: datetime64[ns](1), float64(61), int64(1), object(4)
memory usage: 219.5+ MB
```

In [75]:

```python
#Tramite requests posso recuperare i metadati dalla fonte in github usando
import requests

try:
    dataset_readme = requests.get('https://raw.githubusercontent.com/owid/
except:
    dataset_readme = meta_info = ''

print("METADATI:")
for meta in sorted(df_covid.columns.values):
    if dataset_readme:
        meta_info = dataset_readme[dataset_readme.index(f'{meta}'):]
        meta_info = meta_info[meta_info.index('|') + 1:]
        meta_info = ' -->' + meta_info[:meta_info.index('|') - 1]

    print('-'*len(meta))
    print(f'{meta}{meta_info}')
```

```
METADATI:
-------------
aged_65_older --> Share of the population that is 65 years and older, most re
cent year available
-------------
aged_70_older --> Share of the population that is 70 years and older in 2015
---------------------
cardiovasc_death_rate --> Death rate from cardiovascular disease in 2017 (ann
ual number of deaths per 100,000 people)
---------
continent -->
----
date --> [XLSX](https://covid.ourworldindata.org/data/owid-covid-data.xlsx)
-------------------
diabetes_prevalence --> Diabetes prevalence (% of population aged 20 to 79) i
n 2017
----------------
excess_mortality --> Percentage difference between the reported number of wee
kly or monthly deaths in 2020–2021 and the projected number of deaths for the
same period based on previous years. For more information, see https://githu
b.com/owid/covid-19-data/tree/master/public/data/excess_mortality
---------------------------
excess_mortality_cumulative --> Percentage difference between the cumulative
number of deaths since 1 January 2020 and the cumulative projected deaths for
the same period based on previous years. For more information, see https://gi
thub.com/owid/covid-19-data/tree/master/public/data/excess_mortality
------------------------------------
excess_mortality_cumulative_absolute --> Cumulative difference between the re
ported number of deaths since 1 January 2020 and the projected number of deat
hs for the same period based on previous years. For more information, see htt
ps://github.com/owid/covid-19-data/tree/master/public/data/excess_mortality
---------------------------------------
```

```
excess_mortality_cumulative_per_million --> Cumulative difference between the
reported number of deaths since 1 January 2020 and the projected number of de
aths for the same period based on previous years, per million people. For mor
e information, see https://github.com/owid/covid-19-data/tree/master/public/d
ata/excess_mortality
---------------
extreme_poverty --> Share of the population living in extreme poverty, most r
ecent year available since 2010
--------------
female_smokers --> Share of women who smoke, most recent year available
--------------
gdp_per_capita --> Gross domestic product at purchasing power parity (constan
t 2011 international dollars), most recent year available
---------------------
handwashing_facilities --> Share of the population with basic handwashing fac
ilities on premises, most recent year available
-------------
hosp_patients --> Number of COVID-19 patients in hospital on a given day
------------------------
hosp_patients_per_million --> Number of COVID-19 patients in hospital on a gi
ven day per 1,000,000 people
-------------------------
hospital_beds_per_thousand --> Hospital beds per 1,000 people, most recent ye
ar available since 2010
-----------------------
human_development_index --> A composite index measuring average achievement i
n three basic dimensions of human development—a long and healthy life, knowle
dge and a decent standard of living. Values for 2019, imported from http://hd
r.undp.org/en/indicators/137506
------------
icu_patients --> Number of COVID-19 patients in intensive care units (ICUs) o
n a given day
------------------------
icu_patients_per_million --> Number of COVID-19 patients in intensive care un
its (ICUs) on a given day per 1,000,000 people
--------
iso_code --> ISO 3166-1 alpha-3 – three-letter country codes. Note that OWID-
defined regions (e.g. continents like 'Europe') contain prefix 'OWID_'.
---------------
life_expectancy --> Life expectancy at birth in 2019
--------
location --> Variable
------------
male_smokers --> Share of women who smoke, most recent year available
----------
median_age --> Median age of the population, UN projection for 2020
---------
new_cases --> New confirmed cases of COVID-19. Counts can include probable ca
ses, where reported. In rare cases where our source reports a negative daily
change due to a data correction, we set this metric to NA.
---------------------
new_cases_per_million --> New confirmed cases of COVID-19 per 1,000,000 peopl
e. Counts can include probable cases, where reported.
------------------
new_cases_smoothed --> New confirmed cases of COVID-19 (7-day smoothed). Coun
ts can include probable cases, where reported.
-----------------------------
new_cases_smoothed_per_million --> New confirmed cases of COVID-19 (7-day smo
othed) per 1,000,000 people. Counts can include probable cases, where reporte
d.
----------
new_deaths --> New deaths attributed to COVID-19. Counts can include probable
deaths, where reported. In rare cases where our source reports a negative dai
ly change due to a data correction, we set this metric to NA.
```

```
----------------------
new_deaths_per_million --> New deaths attributed to COVID-19 per 1,000,000 pe
ople. Counts can include probable deaths, where reported.
-------------------
new_deaths_smoothed --> New deaths attributed to COVID-19 (7-day smoothed). C
ounts can include probable deaths, where reported.
-------------------------------
new_deaths_smoothed_per_million --> New deaths attributed to COVID-19 (7-day
smoothed) per 1,000,000 people. Counts can include probable deaths, where rep
orted.
------------------------------
new_people_vaccinated_smoothed --> Daily number of people receiving their fir
st vaccine dose (7-day smoothed)
------------------------------------------
new_people_vaccinated_smoothed_per_hundred --> Daily number of people receivi
ng their first vaccine dose (7-day smoothed) per 100 people in the total popu
lation
---------
new_tests --> New tests for COVID-19 (only calculated for consecutive days)
----------------------
new_tests_per_thousand --> New tests for COVID-19 per 1,000 people
------------------
new_tests_smoothed --> New tests for COVID-19 (7-day smoothed). For countries
that don't report testing data on a daily basis, we assume that testing chang
ed equally on a daily basis over any periods in which no data was reported. T
his produces a complete series of daily figures, which is then averaged over
a rolling 7-day window
-------------------------------
new_tests_smoothed_per_thousand --> New tests for COVID-19 (7-day smoothed) p
er 1,000 people
----------------
new_vaccinations --> New COVID-19 vaccination doses administered (only calcul
ated for consecutive days)
-------------------------
new_vaccinations_smoothed --> New COVID-19 vaccination doses administered (7-
day smoothed). For countries that don't report vaccination data on a daily ba
sis, we assume that vaccination changed equally on a daily basis over any per
iods in which no data was reported. This produces a complete series of daily
figures, which is then averaged over a rolling 7-day window
-------------------------------------
new_vaccinations_smoothed_per_million --> New COVID-19 vaccination doses admi
nistered (7-day smoothed) per 1,000,000 people in the total population
----------------------
people_fully_vaccinated --> Total number of people who received all doses pre
scribed by the initial vaccination protocol
-----------------------------------
people_fully_vaccinated_per_hundred --> Total number of people who received a
ll doses prescribed by the initial vaccination protocol per 100 people in the
total population
-----------------
people_vaccinated --> Total number of people who received at least one vaccin
e dose
-----------------------------
people_vaccinated_per_hundred --> Total number of people who received at leas
t one vaccine dose per 100 people in the total population
----------
population -->
------------------
population_density --> Number of people divided by land area, measured in squ
are kilometers, most recent year available
-------------
positive_rate --> The share of COVID-19 tests that are positive, given as a r
olling 7-day average (this is the inverse of tests_per_case)
-----------------
reproduction_rate --> Real-time estimate of the effective reproduction rate
```

```
(R) of COVID-19. See https://github.com/crondonm/TrackingR/tree/main/Estimate
s-Database
----------------
stringency_index --> Government Response Stringency Index: composite measure
based on 9 response indicators including school closures, workplace closures,
and travel bans, rescaled to a value from 0 to 100 (100 = strictest response)
--------------
tests_per_case -->
-----------
tests_units --> Units used by the location to report its testing data. A coun
try file can't contain mixed units. All metrics concerning testing data use t
he specified test unit. Valid units are 'people tested' (number of people tes
ted), 'tests performed' (number of tests performed. a single person can be te
sted more than once in a given day) and 'samples tested' (number of samples t
ested. In some cases, more than one sample may be required to perform a given
test.)
--------------
total_boosters --> Total number of COVID-19 vaccination booster doses adminis
tered (doses administered beyond the number prescribed by the vaccination pro
tocol)
-------------------------
total_boosters_per_hundred --> Total number of COVID-19 vaccination booster d
oses administered per 100 people in the total population
-----------
total_cases --> Total confirmed cases of COVID-19. Counts can include probabl
e cases, where reported.
----------------------
total_cases_per_million --> Total confirmed cases of COVID-19 per 1,000,000 p
eople. Counts can include probable cases, where reported.
------------
total_deaths --> Total deaths attributed to COVID-19. Counts can include prob
able deaths, where reported.
-----------------------
total_deaths_per_million --> Total deaths attributed to COVID-19 per 1,000,00
0 people. Counts can include probable deaths, where reported.
-----------
total_tests --> Total tests for COVID-19
-----------------------
total_tests_per_thousand --> Total tests for COVID-19 per 1,000 people
-----------------
total_vaccinations --> Total number of COVID-19 vaccination doses administere
d
-----------------------------
total_vaccinations_per_hundred --> Total number of COVID-19 vaccination doses
administered per 100 people in the total population
---------------------
weekly_hosp_admissions --> Number of COVID-19 patients newly admitted to hosp
itals in a given week (reporting date and the preceeding 6 days)
---------------------------------
weekly_hosp_admissions_per_million --> Number of COVID-19 patients newly admi
tted to hospitals in a given week per 1,000,000 people (reporting date and th
e preceeding 6 days)
---------------------
weekly_icu_admissions --> Number of COVID-19 patients newly admitted to inten
sive care units (ICUs) in a given week (reporting date and the preceeding 6 d
ays)
--------------------------------
weekly_icu_admissions_per_million --> Number of COVID-19 patients newly admit
ted to intensive care units (ICUs) in a given week per 1,000,000 people (repo
rting date and the preceeding 6 days)
```

In [76]:

```
# Analizzo i descrittori statistici di base per le colonne numeriche

# Seleziono solo le colonne numeriche
```

```
colonne_numeriche = df_covid.select_dtypes(include=['float64', 'int64']).c

# Ne Calcolo i descrittori
df_covid_describe = df_covid[colonne_numeriche].describe().transpose().ast

for col in df_covid_describe.index: #Con questo ciclo for è possibile stam
  n_null = df_covid[col].isnull().sum() #Calcolo valori nulli
  perc_null = df_covid[col].isnull().mean() * 100 #Calcolo percentuale nul
  print(f"Statistiche per la colonna:\n'{col}'")
  df_covid_describe.loc[col, 'valori nulli'] = n_null #aggiungo i valori n
  df_covid_describe.loc[col, '% valori nulli'] = perc_null
  display(df_covid_describe.loc[[col]].transpose())  # il doppio [[]] mant
  print()
```

```
Statistiche per la colonna:
'total_cases'
```

	total_cases
count	4.118040e+05
mean	7.365292e+06
std	4.477582e+07
min	0.000000e+00
25%	6.280750e+03
50%	6.365300e+04
75%	7.582720e+05
max	7.758668e+08
valori nulli	1.763100e+04
% valori nulli	4.105627e+00

```
Statistiche per la colonna:
'new_cases'
```

	new_cases
count	4.101590e+05
mean	8.017360e+03
std	2.296649e+05
min	0.000000e+00
25%	0.000000e+00
50%	0.000000e+00
75%	0.000000e+00
max	4.423623e+07
valori nulli	1.927600e+04
% valori nulli	4.488689e+00

```
Statistiche per la colonna:
'new_cases_smoothed'
```

'new_cases_smoothed'

	new_cases_smoothed
count	4.089290e+05
mean	8.041026e+03
std	8.661611e+04
min	0.000000e+00
25%	0.000000e+00
50%	1.200000e+01
75%	3.132860e+02
max	6.319461e+06
valori nulli	2.050600e+04
% valori nulli	4.775111e+00

```
Statistiche per la colonna:
'total_deaths'
```

	total_deaths
count	4.118040e+05
mean	8.125957e+04
std	4.411901e+05
min	0.000000e+00
25%	4.300000e+01
50%	7.990000e+02
75%	9.574000e+03
max	7.057132e+06
valori nulli	1.763100e+04
% valori nulli	4.105627e+00

```
Statistiche per la colonna:
'new_deaths'
```

	new_deaths
count	410608.000000
mean	71.852139
std	1368.322990
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	103719.000000
valori nulli	18827.000000

valori nulli	18827.000000
% valori nulli	4.384133

```
Statistiche per la colonna:
'new_deaths_smoothed'
```

	new_deaths_smoothed
count	409378.000000
mean	72.060873
std	513.636567
min	0.000000
25%	0.000000
50%	0.000000
75%	3.143000
max	14817.000000
valori nulli	20057.000000
% valori nulli	4.670555

```
Statistiche per la colonna:
'total_cases_per_million'
```

	total_cases_per_million
count	411804.000000
mean	112096.199396
std	162240.412419
min	0.000000
25%	1916.100500
50%	29145.475000
75%	156770.190000
max	763598.600000
valori nulli	17631.000000
% valori nulli	4.105627

```
Statistiche per la colonna:
'new_cases_per_million'
```

	new_cases_per_million
count	410159.000000
mean	122.357074
std	1508.778583
min	0.000000
25%	0.000000
50%	0.000000

75%	0.000000
max	241758.230000
valori nulli	19276.000000
% valori nulli	4.488689

```
Statistiche per la colonna:
'new_cases_smoothed_per_million'
```

	new_cases_smoothed_per_million
count	408929.000000
mean	122.713844
std	559.701638
min	0.000000
25%	0.000000
50%	2.794000
75%	56.253000
max	34536.890000
valori nulli	20506.000000
% valori nulli	4.775111

```
Statistiche per la colonna:
'total_deaths_per_million'
```

	total_deaths_per_million
count	411804.000000
mean	835.514313
std	1134.932671
min	0.000000
25%	24.568000
50%	295.089000
75%	1283.817000
max	6601.110000
valori nulli	17631.000000
% valori nulli	4.105627

```
Statistiche per la colonna:
'new_deaths_per_million'
```

	new_deaths_per_million
count	410608.000000
mean	0.762323
std	6.982537

min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	893.655000
valori nulli	18827.000000
% valori nulli	4.384133

```
Statistiche per la colonna:
'new_deaths_smoothed_per_million'
```

	new_deaths_smoothed_per_million
count	409378.000000
mean	0.764555
std	2.546519
min	0.000000
25%	0.000000
50%	0.000000
75%	0.357000
max	127.665000
valori nulli	20057.000000
% valori nulli	4.670555

```
Statistiche per la colonna:
'reproduction_rate'
```

	reproduction_rate
count	184817.000000
mean	0.911495
std	0.399925
min	-0.070000
25%	0.720000
50%	0.950000
75%	1.140000
max	5.870000
valori nulli	244618.000000
% valori nulli	56.962753

```
Statistiche per la colonna:
'icu_patients'
```

	icu_patients

count	39116.000000
mean	660.971418
std	2139.615532
min	0.000000
25%	21.000000
50%	90.000000
75%	413.000000
max	28891.000000
valori nulli	390319.000000
% valori nulli	90.891287

```
Statistiche per la colonna:
'icu_patients_per_million'
```

	icu_patients_per_million
count	39116.000000
mean	15.656340
std	22.785489
min	0.000000
25%	2.328000
50%	6.434000
75%	18.779250
max	180.675000
valori nulli	390319.000000
% valori nulli	90.891287

```
Statistiche per la colonna:
'hosp_patients'
```

	hosp_patients
count	40656.000000
mean	3911.741563
std	9845.750485
min	0.000000
25%	186.000000
50%	776.000000
75%	3051.000000
max	154497.000000
valori nulli	388779.000000
% valori nulli	90.532677

```
Statistiche per la colonna:
'hosp_patients_per_million'
```

	hosp_patients_per_million
count	40656.000000
mean	125.988007
std	151.155812
min	0.000000
25%	30.997000
50%	74.236000
75%	159.758250
max	1526.846000
valori nulli	388779.000000
% valori nulli	90.532677

```
Statistiche per la colonna:
'weekly_icu_admissions'
```

	weekly_icu_admissions
count	10993.000000
mean	317.894114
std	514.412910
min	0.000000
25%	17.000000
50%	92.000000
75%	353.000000
max	4838.000000
valori nulli	418442.000000
% valori nulli	97.440125

```
Statistiche per la colonna:
'weekly_icu_admissions_per_million'
```

	weekly_icu_admissions_per_million
count	10993.000000
mean	9.671944
std	13.574017
min	0.000000
25%	1.549000
50%	4.645000
75%	12.651000
max	224.976000

max	224.976000
valori nulli	418442.000000
% valori nulli	97.440125

```
Statistiche per la colonna:
'weekly_hosp_admissions'
```

	weekly_hosp_admissions
count	24497.000000
mean	4291.723313
std	10919.623681
min	0.000000
25%	223.000000
50%	864.000000
75%	3893.000000
max	153977.000000
valori nulli	404938.000000
% valori nulli	94.295528

```
Statistiche per la colonna:
'weekly_hosp_admissions_per_million'
```

	weekly_hosp_admissions_per_million
count	24497.000000
mean	82.619130
std	88.396751
min	0.000000
25%	23.728000
50%	56.277000
75%	109.998000
max	717.077000
valori nulli	404938.000000
% valori nulli	94.295528

```
Statistiche per la colonna:
'total_tests'
```

	total_tests
count	7.938700e+04
mean	2.110457e+07
std	8.409869e+07
min	0.000000e+00
25%	3.646540e+05

50%	2.067330e+06
75%	1.024845e+07
max	9.214000e+09
valori nulli	3.500480e+05
% valori nulli	8.151362e+01

```
Statistiche per la colonna:
'new_tests'
```

	new_tests
count	7.540300e+04
mean	6.728541e+04
std	2.477340e+05
min	1.000000e+00
25%	2.244000e+03
50%	8.783000e+03
75%	3.722900e+04
max	3.585563e+07
valori nulli	3.540320e+05
% valori nulli	8.244135e+01

```
Statistiche per la colonna:
'total_tests_per_thousand'
```

	total_tests_per_thousand
count	79387.000000
mean	924.254762
std	2195.428504
min	0.000000
25%	43.585500
50%	234.141000
75%	894.374500
max	32925.826000
valori nulli	350048.000000
% valori nulli	81.513617

```
Statistiche per la colonna:
'new_tests_per_thousand'
```

	new_tests_per_thousand
count	75403.000000
mean	3.272466

std	9.033843
min	0.000000
25%	0.286000
50%	0.971000
75%	2.914000
max	531.062000
valori nulli	354032.000000
% valori nulli	82.441347

```
Statistiche per la colonna:
'new_tests_smoothed'
```

	new_tests_smoothed
count	1.039650e+05
mean	1.421784e+05
std	1.138215e+06
min	0.000000e+00
25%	1.486000e+03
50%	6.570000e+03
75%	3.220500e+04
max	1.476998e+07
valori nulli	3.254700e+05
% valori nulli	7.579028e+01

```
Statistiche per la colonna:
'new_tests_smoothed_per_thousand'
```

	new_tests_smoothed_per_thousand
count	103965.000000
mean	2.826309
std	7.308233
min	0.000000
25%	0.203000
50%	0.851000
75%	2.584000
max	147.603000
valori nulli	325470.000000
% valori nulli	75.790283

```
Statistiche per la colonna:
'positive_rate'
```

	positive_rate
count	95927.000000
mean	0.098163
std	0.115978
min	0.000000
25%	0.017000
50%	0.055000
75%	0.138100
max	1.000000
valori nulli	333508.000000
% valori nulli	77.662044

```
Statistiche per la colonna:
'tests_per_case'
```

	tests_per_case
count	9.434800e+04
mean	2.403633e+03
std	3.344366e+04
min	1.000000e+00
25%	7.100000e+00
50%	1.750000e+01
75%	5.460000e+01
max	1.023632e+06
valori nulli	3.350870e+05
% valori nulli	7.802974e+01

```
Statistiche per la colonna:
'total_vaccinations'
```

	total_vaccinations
count	8.541700e+04
mean	5.616980e+08
std	1.842160e+09
min	0.000000e+00
25%	1.970788e+06
50%	1.439435e+07
75%	1.161972e+08
max	1.357877e+10
valori nulli	3.440180e+05

% valori nulli	8.010945e+01

```
Statistiche per la colonna:
'people_vaccinated'
```

	people_vaccinated
count	8.113200e+04
mean	2.487064e+08
std	8.006461e+08
min	0.000000e+00
25%	1.050009e+06
50%	6.901088e+06
75%	5.093295e+07
max	5.631264e+09
valori nulli	3.483030e+05
% valori nulli	8.110727e+01

```
Statistiche per la colonna:
'people_fully_vaccinated'
```

	people_fully_vaccinated
count	7.806100e+04
mean	2.286639e+08
std	7.403763e+08
min	1.000000e+00
25%	9.644000e+05
50%	6.191345e+06
75%	4.773185e+07
max	5.177943e+09
valori nulli	3.513740e+05
% valori nulli	8.182239e+01

```
Statistiche per la colonna:
'total_boosters'
```

	total_boosters
count	5.360000e+04
mean	1.505811e+08
std	4.360697e+08
min	1.000000e+00
25%	6.022820e+05
50%	5.765440e+06
75%	4.010073e+07

75%	4.019072e+07
max	2.817381e+09
valori nulli	3.758350e+05
% valori nulli	8.751848e+01

```
Statistiche per la colonna:
'new_vaccinations'
```

	new_vaccinations
count	7.097100e+04
mean	7.398640e+05
std	3.183064e+06
min	0.000000e+00
25%	2.010000e+03
50%	2.053100e+04
75%	1.736115e+05
max	4.967320e+07
valori nulli	3.584640e+05
% valori nulli	8.347340e+01

```
Statistiche per la colonna:
'new_vaccinations_smoothed'
```

	new_vaccinations_smoothed
count	1.950290e+05
mean	2.838758e+05
std	1.922352e+06
min	0.000000e+00
25%	2.790000e+02
50%	3.871000e+03
75%	3.180300e+04
max	4.369181e+07
valori nulli	2.344060e+05
% valori nulli	5.458475e+01

```
Statistiche per la colonna:
'total_vaccinations_per_hundred'
```

	total_vaccinations_per_hundred
count	85417.000000
mean	124.279558
std	85.098042
min	0.000000

25%	44.770000
50%	130.550000
75%	194.990000
max	410.230000
valori nulli	344018.000000
% valori nulli	80.109446

```
Statistiche per la colonna:
'people_vaccinated_per_hundred'
```

	people_vaccinated_per_hundred
count	81132.000000
mean	53.501409
std	29.379655
min	0.000000
25%	27.880000
50%	64.300000
75%	77.780000
max	129.070000
valori nulli	348303.000000
% valori nulli	81.107269

```
Statistiche per la colonna:
'people_fully_vaccinated_per_hundred'
```

	people_fully_vaccinated_per_hundred
count	78061.000000
mean	48.680182
std	29.042282
min	0.000000
25%	21.220000
50%	57.920000
75%	73.610000
max	126.890000
valori nulli	351374.000000
% valori nulli	81.822395

```
Statistiche per la colonna:
'total_boosters_per_hundred'
```

	total_boosters_per_hundred
count	53600.000000

mean	36.301489
std	30.218208
min	0.000000
25%	5.920000
50%	35.905000
75%	57.620000
max	150.470000
valori nulli	375835.000000
% valori nulli	87.518484

```
Statistiche per la colonna:
'new_vaccinations_smoothed_per_million'
```

	new_vaccinations_smoothed_per_million
count	195029.000000
mean	1851.477596
std	3117.828731
min	0.000000
25%	106.000000
50%	605.000000
75%	2402.000000
max	117113.000000
valori nulli	234406.000000
% valori nulli	54.584745

```
Statistiche per la colonna:
'new_people_vaccinated_smoothed'
```

	new_people_vaccinated_smoothed
count	1.921770e+05
mean	1.060707e+05
std	7.866884e+05
min	0.000000e+00
25%	4.300000e+01
50%	7.710000e+02
75%	9.307000e+03
max	2.107127e+07
valori nulli	2.372580e+05
% valori nulli	5.524887e+01

```
Statistiche per la colonna:
```

```
'new_people_vaccinated_smoothed_per_hundred'
```

	new_people_vaccinated_smoothed_per_hundred
count	192177.000000
mean	0.074980
std	0.176216
min	0.000000
25%	0.001000
50%	0.014000
75%	0.073000
max	11.711000
valori nulli	237258.000000
% valori nulli	55.248874

```
Statistiche per la colonna:
'stringency_index'
```

	stringency_index
count	196190.000000
mean	42.877560
std	24.870492
min	0.000000
25%	22.220000
50%	42.850000
75%	62.040000
max	100.000000
valori nulli	233245.000000
% valori nulli	54.314390

```
Statistiche per la colonna:
'population_density'
```

	population_density
count	360492.000000
mean	394.073095
std	1785.451215
min	0.137000
25%	37.728000
50%	88.125000
75%	222.873000
max	20546.766000

valori nulli	68943.000000
% valori nulli	16.054350

```
Statistiche per la colonna:
'median_age'
```

	median_age
count	334663.000000
mean	30.456296
std	9.093554
min	15.100000
25%	22.200000
50%	29.700000
75%	38.700000
max	48.200000
valori nulli	94772.000000
% valori nulli	22.068998

```
Statistiche per la colonna:
'aged_65_older'
```

	aged_65_older
count	323270.000000
mean	8.684103
std	6.093193
min	1.144000
25%	3.526000
50%	6.293000
75%	13.928000
max	27.049000
valori nulli	106165.000000
% valori nulli	24.722018

```
Statistiche per la colonna:
'aged_70_older'
```

	aged_70_older
count	331315.000000
mean	5.486843
std	4.136342
min	0.526000
25%	2.063000

50%	3.871000
75%	8.643000
max	18.493000
valori nulli	98120.000000
% valori nulli	22.848627

```
Statistiche per la colonna:
'gdp_per_capita'
```

	gdp_per_capita
count	328292.000000
mean	18904.182986
std	19829.578099
min	661.240000
25%	4227.630000
50%	12294.876000
75%	27216.445000
max	116935.600000
valori nulli	101143.000000
% valori nulli	23.552575

```
Statistiche per la colonna:
'extreme_poverty'
```

	extreme_poverty
count	211996.000000
mean	13.924729
std	20.073912
min	0.100000
25%	0.600000
50%	2.500000
75%	21.400000
max	77.600000
valori nulli	217439.000000
% valori nulli	50.633740

```
Statistiche per la colonna:
'cardiovasc_death_rate'
```

	cardiovasc_death_rate
count	328865.000000
mean	264.639387
std	120.756836

min	79.370000
25%	175.695000
50%	245.465000
75%	333.436000
max	724.417000
valori nulli	100570.000000
% valori nulli	23.419144

```
Statistiche per la colonna:
'diabetes_prevalence'
```

	diabetes_prevalence
count	345911.000000
mean	8.556055
std	4.934656
min	0.990000
25%	5.350000
50%	7.200000
75%	10.790000
max	30.530000
valori nulli	83524.000000
% valori nulli	19.449742

```
Statistiche per la colonna:
'female_smokers'
```

	female_smokers
count	247165.000000
mean	10.772465
std	10.761080
min	0.100000
25%	1.900000
50%	6.300000
75%	19.300000
max	44.000000
valori nulli	182270.000000
% valori nulli	42.444142

```
Statistiche per la colonna:
'male_smokers'
```

	male_smokers

count	243817.000000
mean	33.097723
std	13.853948
min	7.700000
25%	22.600000
50%	33.100000
75%	41.500000
max	78.100000
valori nulli	185618.000000
% valori nulli	43.223771

```
Statistiche per la colonna:
'handwashing_facilities'
```

	handwashing_facilities
count	161741.000000
mean	50.649264
std	31.905375
min	1.188000
25%	20.859000
50%	49.542000
75%	82.502000
max	100.000000
valori nulli	267694.000000
% valori nulli	62.336326

```
Statistiche per la colonna:
'hospital_beds_per_thousand'
```

	hospital_beds_per_thousand
count	290689.000000
mean	3.106912
std	2.549205
min	0.100000
25%	1.300000
50%	2.500000
75%	4.210000
max	13.800000
valori nulli	138746.000000
% valori nulli	32.308964

```
Statistiche per la colonna:
'life_expectancy'
```

	life_expectancy
count	390299.000000
mean	73.702098
std	7.387914
min	53.280000
25%	69.500000
50%	75.050000
75%	79.460000
max	86.750000
valori nulli	39136.000000
% valori nulli	9.113370

```
Statistiche per la colonna:
'human_development_index'
```

	human_development_index
count	319127.000000
mean	0.722139
std	0.148903
min	0.394000
25%	0.602000
50%	0.740000
75%	0.829000
max	0.957000
valori nulli	110308.000000
% valori nulli	25.686774

```
Statistiche per la colonna:
'population'
```

	population
count	4.294350e+05
mean	1.520336e+08
std	6.975408e+08
min	4.700000e+01
25%	5.237980e+05
50%	6.336393e+06
75%	3.296952e+07

max	7.975105e+09
valori nulli	0.000000e+00
% valori nulli	0.000000e+00

```
Statistiche per la colonna:
'excess_mortality_cumulative_absolute'
```

	excess_mortality_cumulative_absolute
count	1.341100e+04
mean	5.604765e+04
std	1.568691e+05
min	-3.772610e+04
25%	1.765000e+02
50%	6.815199e+03
75%	3.912804e+04
max	1.349776e+06
valori nulli	4.160240e+05
% valori nulli	9.687706e+01

```
Statistiche per la colonna:
'excess_mortality_cumulative'
```

	excess_mortality_cumulative
count	13411.000000
mean	9.766431
std	12.040658
min	-44.230000
25%	2.060000
50%	8.130000
75%	15.160000
max	78.080000
valori nulli	416024.000000
% valori nulli	96.877059

```
Statistiche per la colonna:
'excess_mortality'
```

	excess_mortality
count	13411.000000
mean	10.925353
std	24.560706
min	-95.920000

25%	-1.500000
50%	5.660000
75%	15.575000
max	378.220000
valori nulli	416024.000000
% valori nulli	96.877059

```
Statistiche per la colonna:
'excess_mortality_cumulative_per_million'
```

	excess_mortality_cumulative_per_million
count	13411.000000
mean	1772.666400
std	1991.892769
min	-2936.453100
25%	116.872242
50%	1270.801400
75%	2883.024150
max	10293.515000
valori nulli	416024.000000
% valori nulli	96.877059

In [77]:

```python
#EDA - Analisi delle colonne categoriche

'''con un ciclo for creo una panoramica sui valori unici, sul conteggio di
 conteggio di valori nulli e relativa percentuale, modalità presenti'''

colonne_categoriche = df_covid.columns
for col in colonne_categoriche:
  if df_covid[col].dtype == 'object':
    print(f"\nColonna: '{col}'")
    print(f"Valori unici: {df_covid[col].nunique()}")
    print(f"Valori:\n{df_covid[col].value_counts()}")
    print(f"Valori nulli: {df_covid[col].isnull().sum()} ({df_covid[col].i
    print(f"Modalità presenti: {df_covid[col].unique()}")
```

```
Colonna: 'iso_code'
Valori unici: 255
Valori:
iso_code
OWID_HIC    3026
OWID_EUN    3024
OWID_UMC    3013
OWID_LMC    2983
OWID_LIC    2724
            ...
OWID_SCT    1305
OWID_WLS    1198
```

```
MAC          795
OWID_CYN     691
ESH            1
Name: count, Length: 255, dtype: int64
Valori nulli: 0 (0.00%)
Modalità presenti: ['AFG' 'OWID_AFR' 'ALB' 'DZA' 'ASM' 'AND' 'AGO' 'AIA' 'AT
G' 'ARG' 'ARM'
 'ABW' 'OWID_ASI' 'AUS' 'AUT' 'AZE' 'BHS' 'BHR' 'BGD' 'BRB' 'BLR' 'BEL'
 'BLZ' 'BEN' 'BMU' 'BTN' 'BOL' 'BES' 'BIH' 'BWA' 'BRA' 'VGB' 'BRN' 'BGR'
 'BFA' 'BDI' 'KHM' 'CMR' 'CAN' 'CPV' 'CYM' 'CAF' 'TCD' 'CHL' 'CHN' 'COL'
 'COM' 'COG' 'COK' 'CRI' 'CIV' 'HRV' 'CUB' 'CUW' 'CYP' 'CZE' 'COD' 'DNK'
 'DJI' 'DMA' 'DOM' 'TLS' 'ECU' 'EGY' 'SLV' 'OWID_ENG' 'GNQ' 'ERI' 'EST'
 'SWZ' 'ETH' 'OWID_EUR' 'OWID_EUN' 'FRO' 'FLK' 'FJI' 'FIN' 'FRA' 'GUF'
 'PYF' 'GAB' 'GMB' 'GEO' 'DEU' 'GHA' 'GIB' 'GRC' 'GRL' 'GRD' 'GLP' 'GUM'
 'GTM' 'GGY' 'GIN' 'GNB' 'GUY' 'HTI' 'OWID_HIC' 'HND' 'HKG' 'HUN' 'ISL'
 'IND' 'IDN' 'IRN' 'IRQ' 'IRL' 'IMN' 'ISR' 'ITA' 'JAM' 'JPN' 'JEY' 'JOR'
 'KAZ' 'KEN' 'KIR' 'OWID_KOS' 'KWT' 'KGZ' 'LAO' 'LVA' 'LBN' 'LSO' 'LBR'
 'LBY' 'LIE' 'LTU' 'OWID_LIC' 'OWID_LMC' 'LUX' 'MAC' 'MDG' 'MWI' 'MYS'
 'MDV' 'MLI' 'MLT' 'MHL' 'MTQ' 'MRT' 'MUS' 'MYT' 'MEX' 'FSM' 'MDA' 'MCO'
 'MNG' 'MNE' 'MSR' 'MAR' 'MOZ' 'MMR' 'NAM' 'NRU' 'NPL' 'NLD' 'NCL' 'NZL'
 'NIC' 'NER' 'NGA' 'NIU' 'OWID_NAM' 'PRK' 'MKD' 'OWID_CYN' 'OWID_NIR'
 'MNP' 'NOR' 'OWID_OCE' 'OMN' 'PAK' 'PLW' 'PSE' 'PAN' 'PNG' 'PRY' 'PER'
 'PHL' 'PCN' 'POL' 'PRT' 'PRI' 'QAT' 'REU' 'ROU' 'RUS' 'RWA' 'BLM' 'SHN'
 'KNA' 'LCA' 'MAF' 'SPM' 'VCT' 'WSM' 'SMR' 'STP' 'SAU' 'OWID_SCT' 'SEN'
 'SRB' 'SYC' 'SLE' 'SGP' 'SXM' 'SVK' 'SVN' 'SLB' 'SOM' 'ZAF' 'OWID_SAM'
 'KOR' 'SSD' 'ESP' 'LKA' 'SDN' 'SUR' 'SWE' 'CHE' 'SYR' 'TWN' 'TJK' 'TZA'
 'THA' 'TGO' 'TKL' 'TON' 'TTO' 'TUN' 'TUR' 'TKM' 'TCA' 'TUV' 'UGA' 'UKR'
 'ARE' 'GBR' 'USA' 'VIR' 'OWID_UMC' 'URY' 'UZB' 'VUT' 'VAT' 'VEN' 'VNM'
 'OWID_WLS' 'WLF' 'ESH' 'OWID_WRL' 'YEM' 'ZMB' 'ZWE']

Colonna: 'continent'
Valori unici: 6
Valori:
continent
Africa           95419
Europe           91031
Asia             84199
North America    68638
Oceania          40183
South America    23440
Name: count, dtype: int64
Valori nulli: 26525 (6.18%)
Modalità presenti: ['Asia' nan 'Europe' 'Africa' 'Oceania' 'North America' 'S
outh America']

Colonna: 'location'
Valori unici: 255
Valori:
location
High-income countries            3026
European Union (27)              3024
Upper-middle-income countries    3013
Lower-middle-income countries    2983
Low-income countries             2724
                                 ...
Scotland                         1305
Wales                            1198
Macao                             795
Northern Cyprus                   691
Western Sahara                      1
Name: count, Length: 255, dtype: int64
Valori nulli: 0 (0.00%)
Modalità presenti: ['Afghanistan' 'Africa' 'Albania' 'Algeria' 'American Samo
a' 'Andorra'
 'Angola' 'Anguilla' 'Antigua and Barbuda' 'Argentina' 'Armenia' 'Aruba'
```

```
 'Asia' 'Australia' 'Austria' 'Azerbaijan' 'Bahamas' 'Bahrain'
 'Bangladesh' 'Barbados' 'Belarus' 'Belgium' 'Belize' 'Benin' 'Bermuda'
 'Bhutan' 'Bolivia' 'Bonaire Sint Eustatius and Saba'
 'Bosnia and Herzegovina' 'Botswana' 'Brazil' 'British Virgin Islands'
 'Brunei' 'Bulgaria' 'Burkina Faso' 'Burundi' 'Cambodia' 'Cameroon'
 'Canada' 'Cape Verde' 'Cayman Islands' 'Central African Republic' 'Chad'
 'Chile' 'China' 'Colombia' 'Comoros' 'Congo' 'Cook Islands' 'Costa Rica'
 "Cote d'Ivoire" 'Croatia' 'Cuba' 'Curacao' 'Cyprus' 'Czechia'
 'Democratic Republic of Congo' 'Denmark' 'Djibouti' 'Dominica'
 'Dominican Republic' 'East Timor' 'Ecuador' 'Egypt' 'El Salvador'
 'England' 'Equatorial Guinea' 'Eritrea' 'Estonia' 'Eswatini' 'Ethiopia'
 'Europe' 'European Union (27)' 'Faroe Islands' 'Falkland Islands' 'Fiji'
 'Finland' 'France' 'French Guiana' 'French Polynesia' 'Gabon' 'Gambia'
 'Georgia' 'Germany' 'Ghana' 'Gibraltar' 'Greece' 'Greenland' 'Grenada'
 'Guadeloupe' 'Guam' 'Guatemala' 'Guernsey' 'Guinea' 'Guinea-Bissau'
 'Guyana' 'Haiti' 'High-income countries' 'Honduras' 'Hong Kong' 'Hungary'
 'Iceland' 'India' 'Indonesia' 'Iran' 'Iraq' 'Ireland' 'Isle of Man'
 'Israel' 'Italy' 'Jamaica' 'Japan' 'Jersey' 'Jordan' 'Kazakhstan' 'Kenya'
 'Kiribati' 'Kosovo' 'Kuwait' 'Kyrgyzstan' 'Laos' 'Latvia' 'Lebanon'
 'Lesotho' 'Liberia' 'Libya' 'Liechtenstein' 'Lithuania'
 'Low-income countries' 'Lower-middle-income countries' 'Luxembourg'
 'Macao' 'Madagascar' 'Malawi' 'Malaysia' 'Maldives' 'Mali' 'Malta'
 'Marshall Islands' 'Martinique' 'Mauritania' 'Mauritius' 'Mayotte'
 'Mexico' 'Micronesia (country)' 'Moldova' 'Monaco' 'Mongolia'
 'Montenegro' 'Montserrat' 'Morocco' 'Mozambique' 'Myanmar' 'Namibia'
 'Nauru' 'Nepal' 'Netherlands' 'New Caledonia' 'New Zealand' 'Nicaragua'
 'Niger' 'Nigeria' 'Niue' 'North America' 'North Korea' 'North Macedonia'
 'Northern Cyprus' 'Northern Ireland' 'Northern Mariana Islands' 'Norway'
 'Oceania' 'Oman' 'Pakistan' 'Palau' 'Palestine' 'Panama'
 'Papua New Guinea' 'Paraguay' 'Peru' 'Philippines' 'Pitcairn' 'Poland'
 'Portugal' 'Puerto Rico' 'Qatar' 'Reunion' 'Romania' 'Russia' 'Rwanda'
 'Saint Barthelemy' 'Saint Helena' 'Saint Kitts and Nevis' 'Saint Lucia'
 'Saint Martin (French part)' 'Saint Pierre and Miquelon'
 'Saint Vincent and the Grenadines' 'Samoa' 'San Marino'
 'Sao Tome and Principe' 'Saudi Arabia' 'Scotland' 'Senegal' 'Serbia'
 'Seychelles' 'Sierra Leone' 'Singapore' 'Sint Maarten (Dutch part)'
 'Slovakia' 'Slovenia' 'Solomon Islands' 'Somalia' 'South Africa'
 'South America' 'South Korea' 'South Sudan' 'Spain' 'Sri Lanka' 'Sudan'
 'Suriname' 'Sweden' 'Switzerland' 'Syria' 'Taiwan' 'Tajikistan'
 'Tanzania' 'Thailand' 'Togo' 'Tokelau' 'Tonga' 'Trinidad and Tobago'
 'Tunisia' 'Turkey' 'Turkmenistan' 'Turks and Caicos Islands' 'Tuvalu'
 'Uganda' 'Ukraine' 'United Arab Emirates' 'United Kingdom'
 'United States' 'United States Virgin Islands'
 'Upper-middle-income countries' 'Uruguay' 'Uzbekistan' 'Vanuatu'
 'Vatican' 'Venezuela' 'Vietnam' 'Wales' 'Wallis and Futuna'
 'Western Sahara' 'World' 'Yemen' 'Zambia' 'Zimbabwe']

Colonna: 'tests_units'
Valori unici: 4
Valori:
tests_units
tests performed    80099
people tested      16257
samples tested      9591
units unclear        841
Name: count, dtype: int64
Valori nulli: 322647 (75.13%)
Modalità presenti: [nan 'tests performed' 'units unclear' 'samples tested' 'p
eople tested']
```

2. Si chiede di trovare, per ogni continente:

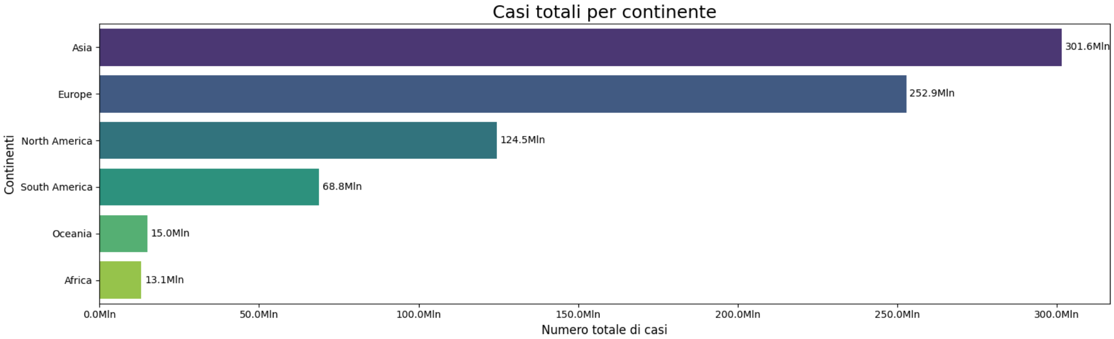
- a. il numero di casi fin dall'inizio della pandemia

- b. la percentuale rispetto al totale mondiale del numero di casi

In [78]:

```
#Eseguo il conteggio sommando i nuovi casi raggruppati per continente
conteggio_casi = df_covid.groupby('continent')['new_cases'].sum().sort_val
print(conteggio_casi)

# barplot
import matplotlib.ticker as mticker
plt.figure(figsize=(16, 5))
ax = sns.barplot(x=conteggio_casi.values,
                 y=conteggio_casi.index,
                 palette='viridis',
                 orient="h",
                 hue=conteggio_casi.index)
plt.title('Casi totali per continente', fontsize=18)
plt.xlabel('Numero totale di casi', fontsize=12)
plt.ylabel('Continenti', fontsize=12)
plt.title('Casi totali per continente', fontsize=18)

# Formatto l'asse x in milioni (es: 252.9M)
ax.xaxis.set_major_formatter(mticker.FuncFormatter(lambda x, _: f'{x/1e6:.

# Aggiunta etichette sulle barre (in milioni)
for i, v in enumerate(conteggio_casi.values):
    ax.text(
        v + 1e6,  # piccolo spostamento a destra
        i,
        f'{v/1e6:.1f}Mln',
        va='center',
        ha='left',
        fontsize=10,

    )

plt.tight_layout()

plt.show()
```

```
continent
Asia             301564180.0
Europe           252916868.0
North America    124492698.0
South America     68811012.0
Oceania           15003468.0
Africa            13146831.0
Name: new_cases, dtype: float64
```



In [79]:

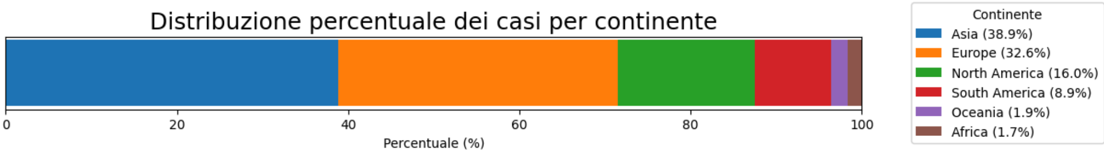
```
#Percentuale rispetto al totale mondiale del numero dei casi
```

```
conteggio_casi_perc = conteggio_casi / conteggio_casi.sum() * 100
print(conteggio_casi_perc)
```

```
continent
Asia             38.864616
Europe           32.595108
North America    16.044216
South America     8.868141
Oceania           1.933598
Africa            1.694321
Name: new_cases, dtype: float64
```

In [80]:

```
import matplotlib.pyplot as plt

#Creo un grafico a barre impilate per visualizzare le proporzioni percentu
valori = conteggio_casi_perc.values
etichette = conteggio_casi_perc.index

plt.figure(figsize=(12, 1.5))
bottom = 0
for i, (val, label) in enumerate(zip(valori, etichette)):
    plt.barh(0, val, left=bottom, label=f"{label} ({val:.1f}%)")
    bottom += val

plt.xlabel('Percentuale (%)')
plt.xlim(0, 100)
plt.yticks([])
plt.title('Distribuzione percentuale dei casi per continente', fontsize=18
plt.legend(loc='center left', bbox_to_anchor=(1.05, 0.5), title="Continent
plt.tight_layout( pad=0.05)
plt.show()
```



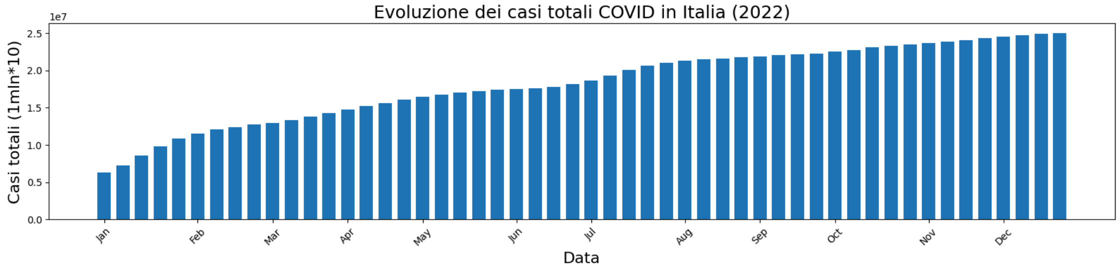
3. Selezionare i dati relativi all'Italia nel 2022 e, poiché i nuovi casi vengono registrati settimanalmente, filtrare via i giorni che non hanno misurazioni; quindi mostrare con dei grafici adeguati:

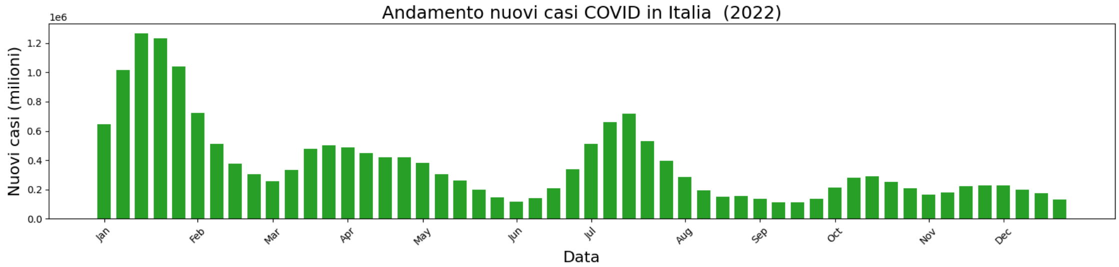
- a. l'evoluzione del casi totali dall'inizio alla fine dell'anno
- b. il numero di nuovi casi rispetto alla data

In [81]:

```
import pandas as pd
import matplotlib.pyplot as plt
import matplotlib.ticker as mticker
import matplotlib.dates as mdates

#Seleziono solo i dati per l'Italia
df_ita = df_covid[df_covid['location'] == 'Italy']
```

```
#Filtro solo il 2022
df_ita_2022 = df_ita[df_ita['date'].dt.year == 2022]

#Elimino i giorni senza misurazione (solo dove ci sono dati sui nuovi casi,
df_ita_2022 = df_ita_2022[df_ita_2022['new_cases'].notnull() & (df_ita_202


# a.Evoluzione dei casi totali
plt.figure(figsize=(16, 4))
plt.bar(df_ita_2022['date'], df_ita_2022['total_cases'], color='tab:blue',
plt.title('Evoluzione dei casi totali COVID in Italia (2022)', fontsize=18
plt.xlabel('Data', fontsize=16)
plt.ylabel('Casi totali (1mln*10)', fontsize=16)

#Per visualizzare le etichette con in nomi dei mesi vanno aggiunte le segu
df_ita_2022['month'] = df_ita_2022['date'].dt.month
df_ita_2022['month_name'] = df_ita_2022['date'].dt.strftime('%b')  # 'Jan'

# Risulta poi necessario trovare le date corrispondenti all'inizio di ogni
month_start_dates = df_ita_2022.groupby(df_ita_2022['date'].dt.to_period('

#Modifica delle etichette sull'asse x
plt.xticks(ticks=month_start_dates, labels=df_ita_2022['month_name'].uniqu
plt.tight_layout()
plt.show()


# b.Numero di nuovi casi per data
plt.figure(figsize=(16, 4))
plt.bar(df_ita_2022['date'], df_ita_2022['new_cases'], width=5,color="tab:
plt.title('Andamento nuovi casi COVID in Italia  (2022)', fontsize=18)
plt.xlabel('Data', fontsize=16)
plt.ylabel('Nuovi casi (milioni)',fontsize=16)
plt.xticks(ticks=month_start_dates, labels=df_ita_2022['month_name'].uniqu
plt.tight_layout()
plt.show()
```





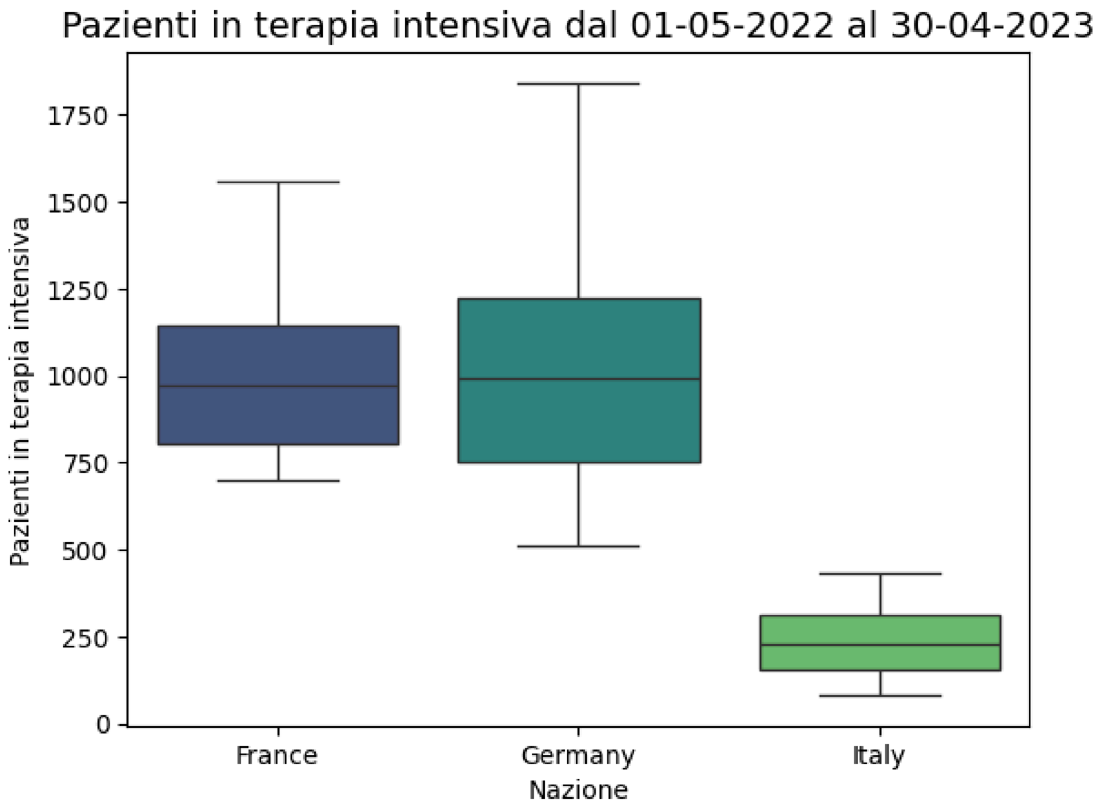
4. Riguardo le nazioni di Italia, Germania e Francia:

- a. mostrare in un boxplot la differenza tra queste nazioni riguardo il numero di pazienti in terapia intensiva (Intensive Care Unit, ICU, considerare quindi la colonna icu_patients) da maggio 2022 (incluso) ad aprile 2023 (incluso)
- b. scrivere un breve commento (una o due righe) riguardo che conclusioni possiamo trarre osservando il grafico risultante

In [82]:

```
#a. Boxplot per confronto numero di pazienti tra Francia, Italia, Germania

df_ita_ger_fra = df_covid[(df_covid['location'] == 'Italy') | (df_covid['l
date_icu = df_ita_ger_fra[(df_ita_ger_fra['date'] >= '2022-05-01') & (df_i
df_ita_ger_fra_icu = date_icu[['location', 'date', 'icu_patients']]
sns.boxplot(x=df_ita_ger_fra_icu["location"],
            y=df_ita_ger_fra_icu["icu_patients"],
            palette="viridis",
            hue=df_ita_ger_fra_icu["location"])
plt.title("Pazienti in terapia intensiva dal 01-05-2022 al 30-04-2023", fo
plt.xlabel("Nazione")
plt.ylabel("Pazienti in terapia intensiva")
```

Out[82]: `Text(0, 0.5, 'Pazienti in terapia intensiva')`



In [83]:

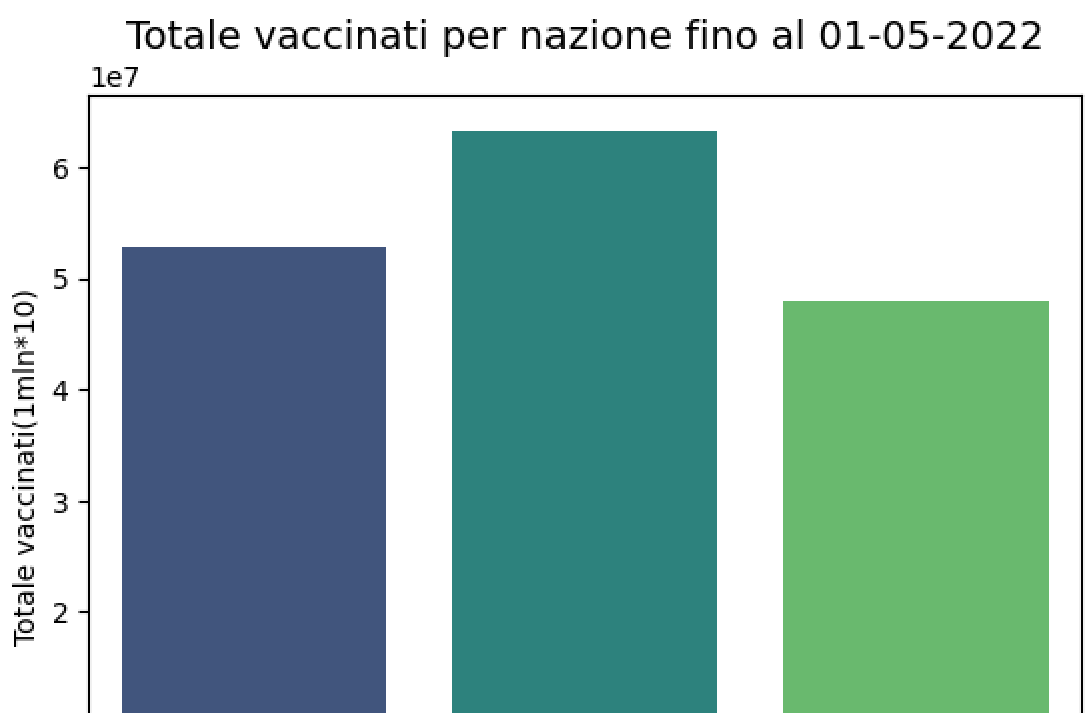
```
#b. Commento sul risultato:

'''Mentre Francia e Germania hanno avuto un numero di pazienti in terapia
l'Italia appare distaccarsi da questo trend, avendo un numero decisamente

Resta però da capire se ciò può essere dovuto a qualche causa o evento spe
Si faranno alcune analisi per valutare delle ipotesi per spiegare questo e

'''
```

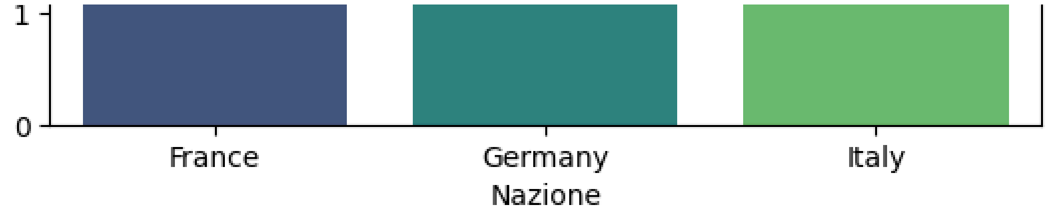
Out[83]:
```
"Mentre Francia e Germania hanno avuto un numero di pazienti in terapia int
ensiva molto simile,\nl'Italia appare distaccarsi da questo trend, avendo u
n numero decisamente inferiore.\n\nResta però da capire se ciò può essere d
ovuto a qualche causa o evento specifici che hanno prodotto questo risultat
o nel periodo analizzato.\nSi faranno alcune analisi per valutare delle ipo
tesi per spiegare questo evento.\n\n"
```

In [84]:
```python
'''Ipotesi 1: Il numero totale di vaccinati ha inciso sui numeri ICU?'''

df_date_vac = df_ita_ger_fra.loc[df_covid['date'] <= '2022-05-01'].copy()
df_ita_ger_fra_spa_vac = df_date_vac[['location', 'date', 'people_fully_va

df_ita_ger_fra_spa_vac = df_ita_ger_fra_spa_vac.groupby('location').max()
display(df_ita_ger_fra_spa_vac)

#Barplot
sns.barplot(x=df_ita_ger_fra_spa_vac.index,
            y=df_ita_ger_fra_spa_vac['people_fully_vaccinated'],
            palette="viridis",
            hue=df_ita_ger_fra_spa_vac.index)
plt.title("Totale vaccinati per nazione fino al 01-05-2022", fontsize=14)
plt.xlabel("Nazione")
plt.ylabel("Totale vaccinati(1mln*10)")

print('''\nIl numero totale per ciascuna nazione sembra comparabile, nonos
```

location	date	people_fully_vaccinated
France	2022-05-01	52852013.0
Germany	2022-05-01	63258946.0
Italy	2022-05-01	47856499.0

```
Il numero totale per ciascuna nazione sembra comparabile, nonostante una nume
ro più alto di vaccinati in Germania. Ciò non sembra supportare l'ipotesi.
```

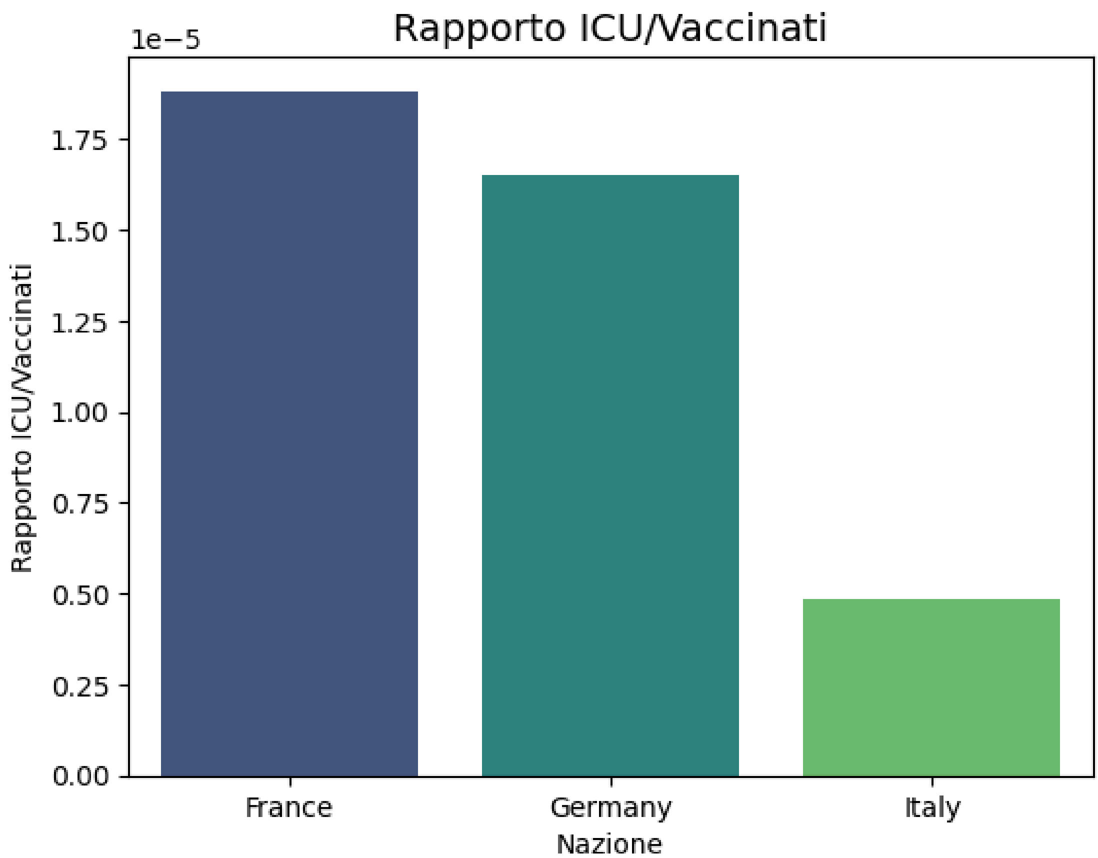




In [85]:

```
'''Rapporto pazienti ICU e numerosità dei vaccinati.'''

rapporto_icu_vac = df_ita_ger_fra_icu['icu_patients']/date_icu['people_ful
rapporto_icu_vac.groupby(date_icu['location']).mean()

#Barplot
sns.barplot(x = rapporto_icu_vac.groupby(date_icu['location']).mean().inde
            y = rapporto_icu_vac.groupby(date_icu['location']).mean().valu
            palette="viridis",
            hue=rapporto_icu_vac.groupby(date_icu['location']).mean().inde
plt.title("Rapporto ICU/Vaccinati", fontsize=14)
plt.xlabel("Nazione")
plt.ylabel("Rapporto ICU/Vaccinati")

print('''\nAnche indagando il rapporto ICU/vaccinati non sembra emergere u
```

```
Anche indagando il rapporto ICU/vaccinati non sembra emergere una spiegazione
sulla bassa numerosità in Italia di ICU nel periodo di riferimento
```



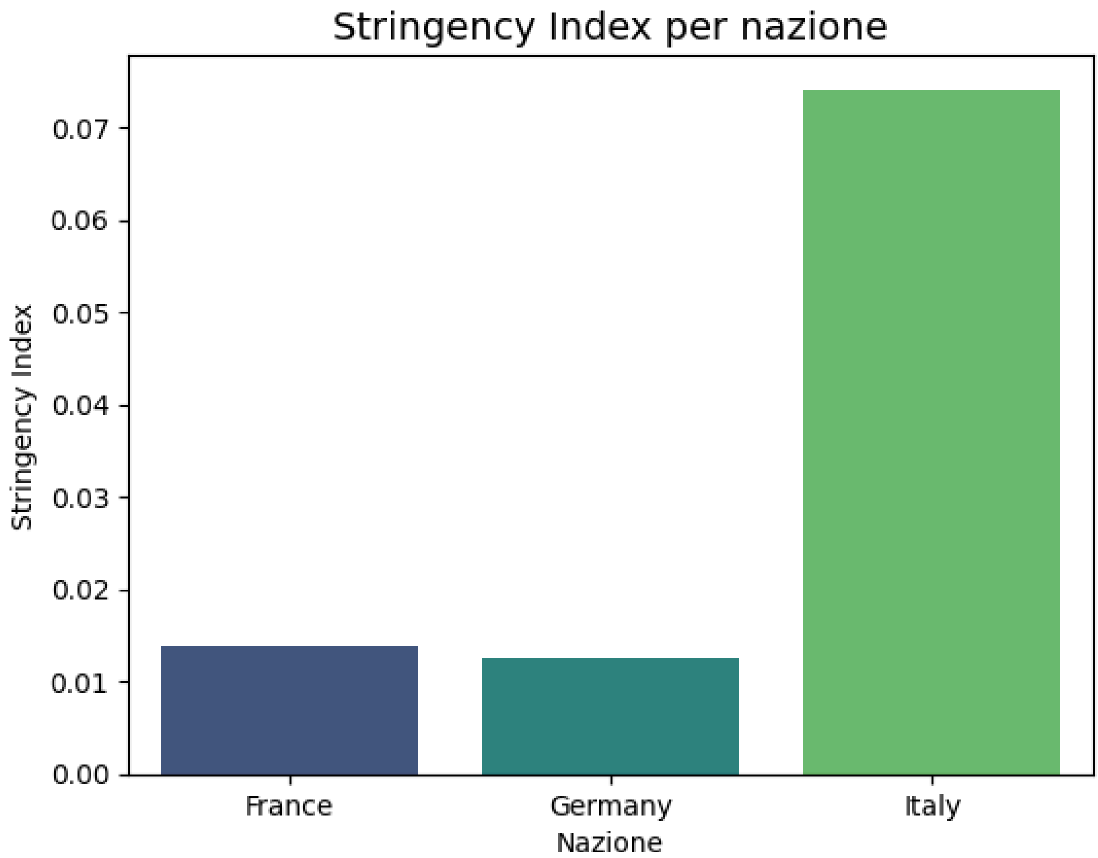
In [86]:

```
'''1. ICU Admission Rate: è possibile che in Italia ci sia stata maggiore
2. ICU Utilization Rate: può il risultato essere dovuto a una distorsione
3. Death to ICU: il risultato potrebbe dipendere dal basso numero di posti
4. Stringency to ICU: può la severità delle misure restrittive messe in at

icu_admission_rate = date_icu['icu_patients'] / date_icu['hosp_patients']
```

```
icu_utilization_rate = date_icu['icu_patients'] / date_icu['population'] #

death_to_icu = date_icu['icu_patients'] / date_icu['total_deaths'] #Death

stringency_to_icu = date_icu['stringency_index'] / date_icu['icu_patients'

print("\nICU Admission Rate\n", icu_admission_rate.groupby(date_icu['locat
print("\nICU Utilization Rate\n", icu_utilization_rate.groupby(date_icu['l
print("\nDeath to ICU\n", death_to_icu.groupby(date_icu['location']).mean(
print("\nStringency to ICU\n", stringency_to_icu.groupby(date_icu['locatio

#Barplot
sns.barplot(x = stringency_to_icu.groupby(date_icu['location']).mean().ind
            y = stringency_to_icu.groupby(date_icu['location']).mean().val
            palette="viridis", hue=stringency_to_icu.groupby(date_icu['loc
plt.title("Stringency Index per nazione", fontsize=14)
plt.xlabel("Nazione")
plt.ylabel("Stringency Index")

#Commento
print('''\nDall’analisi emergono tre evidenze principali:
1) Francia e Germania hanno gestito un carico ICU più elevato, con maggior
2) l’Italia, pur con minor carico ICU e ospedaliero, ha registrato un tass
3) l’approccio italiano è stato il più restrittivo in proporzione ai ricov
L’assenza del dato tedesco sull’ICU admission rate limita una piena interp

Importante notare dal grafico l'ampiezza maggiore dello stringency index i
```

```
ICU Admission Rate
 location
France     0.057915
Germany         NaN
Italy      0.038233
dtype: float64

ICU Utilization Rate
 location
France     0.000015
Germany    0.000012
Italy      0.000004
dtype: float64

Death to ICU
 location
France     0.006388
Germany    0.006507
Italy      0.001311
dtype: float64

Stringency to ICU
 location
France     0.013856
Germany    0.012475
Italy      0.074097
dtype: float64

Dall’analisi emergono tre evidenze principali:
1) Francia e Germania hanno gestito un carico ICU più elevato, con maggiore a
ccesso ai trattamenti intensivi
2) l’Italia, pur con minor carico ICU e ospedaliero, ha registrato un tasso d
```

```
ecessi/ICU più alto, che potrebbe riflettere saturazione o altri fattori di r
ischio
3) l’approccio italiano è stato il più restrittivo in proporzione ai ricoveri
intensivi, suggerendo una strategia maggiormente orientata alla prevenzione a
nticipata
L’assenza del dato tedesco sull’ICU admission rate limita una piena interpret
azione del quadro clinico.

Importante notare dal grafico l'ampiezza maggiore dello stringency index in I
talia rispetto ai valori di Francia e Germania.
```



5. Riguardo le nazioni di Italia, Germania, Francia e Spagna in tutto il 2021:

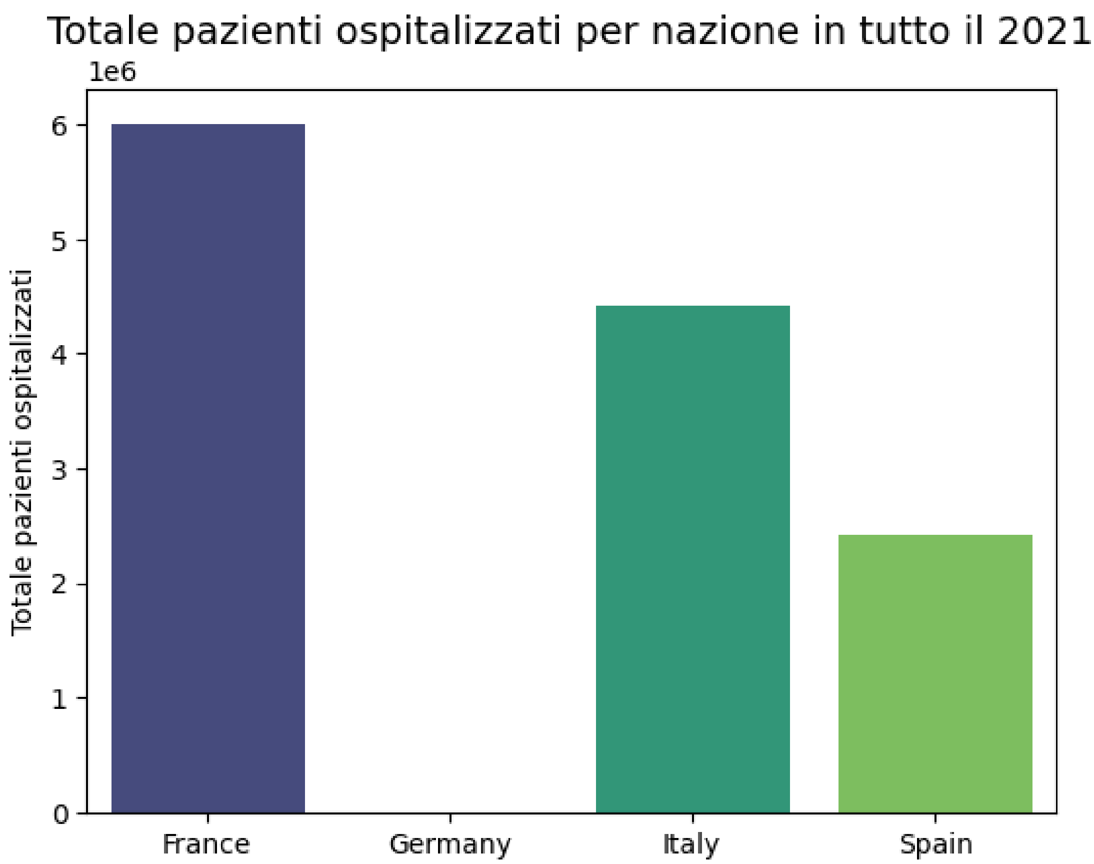
- a. mostrare, in maniera grafica oppure numerica, la somma dei pazienti ospitalizzati per ognuna (colonna hosp_patients)
- b. se ci sono dati nulli, con un breve commento scrivere se può essere possibile gestirli tramite sostituzione o meno

In [87]:

```
#Filtro le nazioni desiderate
df_ita_ger_fra_spa = df_covid[(df_covid['location'] == 'Italy') | (df_covi

#Filtro per l'anno 2021 sulla colonna 'date'
df_ita_ger_fra_spa_2021 = df_ita_ger_fra_spa[df_ita_ger_fra_spa['date'].dt

#Seleziono e raggruppo le colonne
df_ita_ger_fra_spa_hosp = df_ita_ger_fra_spa_2021[['location', 'hosp_patie
df_ita_ger_fra_spa_hosp = df_ita_ger_fra_spa_hosp.groupby('location').sum(

#Risultati e Barplot
print(df_ita_ger_fra_spa_hosp)
sns.barplot(x=df_ita_ger_fra_spa_hosp.index, y=df_ita_ger_fra_spa_hosp['ho
```

```
plt.title("Totale pazienti ospitalizzati per nazione in tutto il 2021", fo
plt.xlabel("Nazione")
plt.ylabel("Totale pazienti ospitalizzati")

#Verifica dati nulli
print("\nVerifica dati nulli nella colonna 'hosp_patients' per le nazioni
print(df_ita_ger_fra_spa_2021.isnull().sum()['hosp_patients'])

# Commento sulla gestione dei dati nulli
print("""
I dati nulli risultanti riguardano la totalità delle osservazioni in Germa
In questo caso non è posibile effettuare sostituzioni, data la totale asse
Si ritiene dunque necessario eliminare la Germania da questa analisi,
almeno sino a quando il dato non potrà essere recuperato da ulteriori font
""")
```

```
          hosp_patients
location
France        6008717.0
Germany             0.0
Italy         4419950.0
Spain         2411706.0

Verifica dati nulli nella colonna 'hosp_patients' per le nazioni selezionate
nel 2021:
365

I dati nulli risultanti riguardano la totalità delle osservazioni in Germania
per il 2021.
In questo caso non è posibile effettuare sostituzioni, data la totale assenza
di misurazioni.
Si ritiene dunque necessario eliminare la Germania da questa analisi,
almeno sino a quando il dato non potrà essere recuperato da ulteriori fonti e
sterne (recuperabile da hosp_admissions).
```

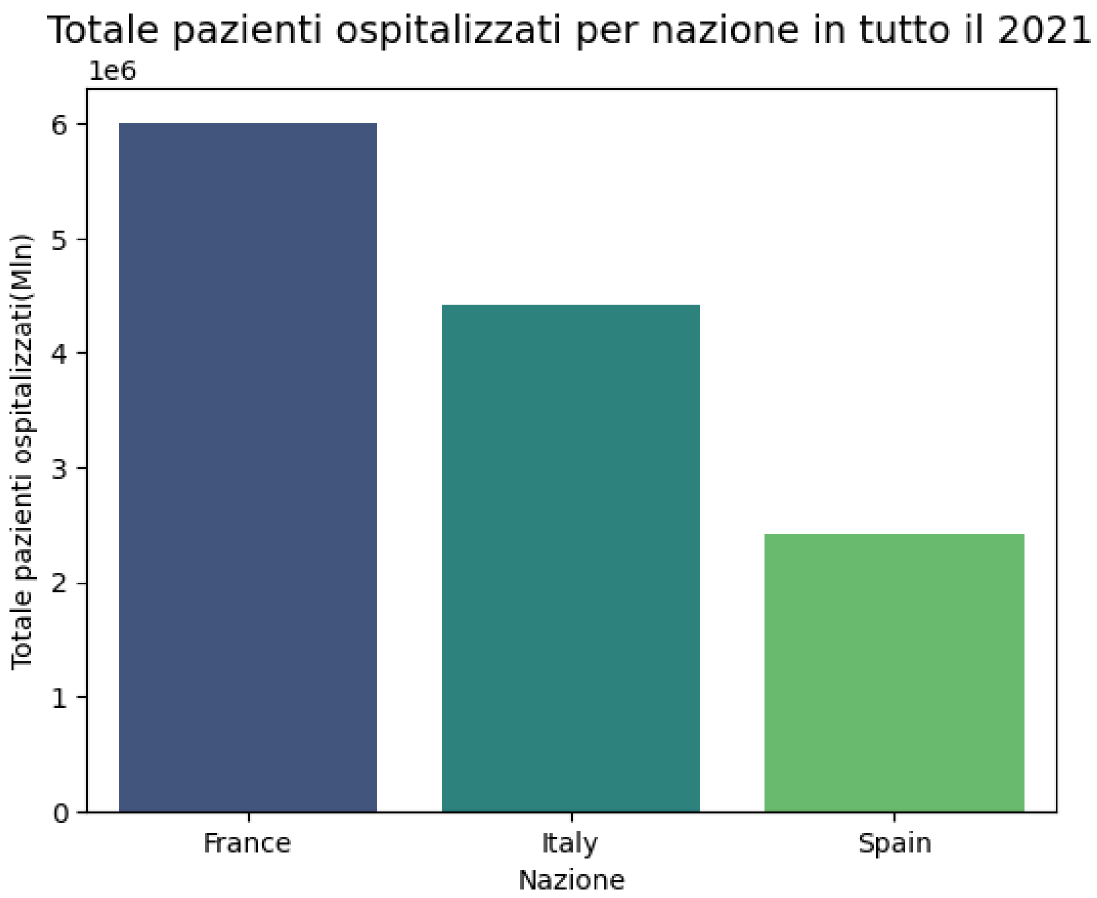


Nazione

In [88]:

```
#Barplot con Germania esclusa
df_ita_fra_spa_2021 = df_ita_ger_fra_spa_2021.drop(df_ita_ger_fra_spa_2021
df_ita_fra_spa_2021_hosp = df_ita_fra_spa_2021[['location', 'hosp_patients
df_ita_fra_spa_2021_hosp = df_ita_fra_spa_2021_hosp.groupby('location').su

sns.barplot(x=df_ita_fra_spa_2021_hosp.index, y=df_ita_fra_spa_2021_hosp['
plt.title("Totale pazienti ospitalizzati per nazione in tutto il 2021", fo
plt.xlabel("Nazione")
plt.ylabel("Totale pazienti ospitalizzati(Mln)")
```

Out[88]: `Text(0, 0.5, 'Totale pazienti ospitalizzati(Mln)')`



In [89]:

```
'''Utilizzando le ammissioni ospedaliere per settimana (weekly_hosp_admiss
si può avere una stima delle proporzioni del numero di ospedalizzazioni to

#Seleziono e raggruppo le colonne
df_ita_ger_fra_spa_adm= df_ita_ger_fra_spa_2021[['location', 'weekly_hosp_
df_ita_ger_fra_spa_adm = df_ita_ger_fra_spa_adm.groupby('location').sum()

#Risultati e Barplot
print(df_ita_ger_fra_spa_adm)
sns.barplot(x=df_ita_ger_fra_spa_adm.index, y=df_ita_ger_fra_spa_adm['week
plt.title("Totale amissioni ospedaliere settimanali per nazione (2021)", f
plt.xlabel("Nazione")
plt.ylabel("Totale pazienti ospitalizzati")

#Verifica dati nulli
```